

MadePlus Smart Shoe – Walking Session Report

1. Session Overview - Total distance: 239 m - Duration: 147 s ($\approx$ 2 min 27 s) - Average pace (including pauses): 10 min 15 s / km - Total steps: 59 - Average stride length written in CSV: 4.052 m (this is distance/steps, a session-average value).

2. Cadence Behaviour - 0–10 s: Cadence is initially 0, then rises into the 20–30 range as you start moving. This is the warm-up phase. - 10–60 s: Cadence stabilises mostly between 30–45. This corresponds to a comfortable, steady walk – neither shuffling nor power walking. - 60–95 s: Cadence gradually dips into the 20–30 band, with a brief low/near-zero patch around $\sim$ 90 s. This looks like slowing down with a small micro-pause or turn. - 112–121 s: Clear fast segment. Cadence jumps into roughly the 55–75 band with a maximum around 80. This is your “bit fast” phase – a short burst of brisk walking. - 121–147 s: Cadence drops again, with a few slower steps and then almost no movement at the very end. This is your cool-down and stop.

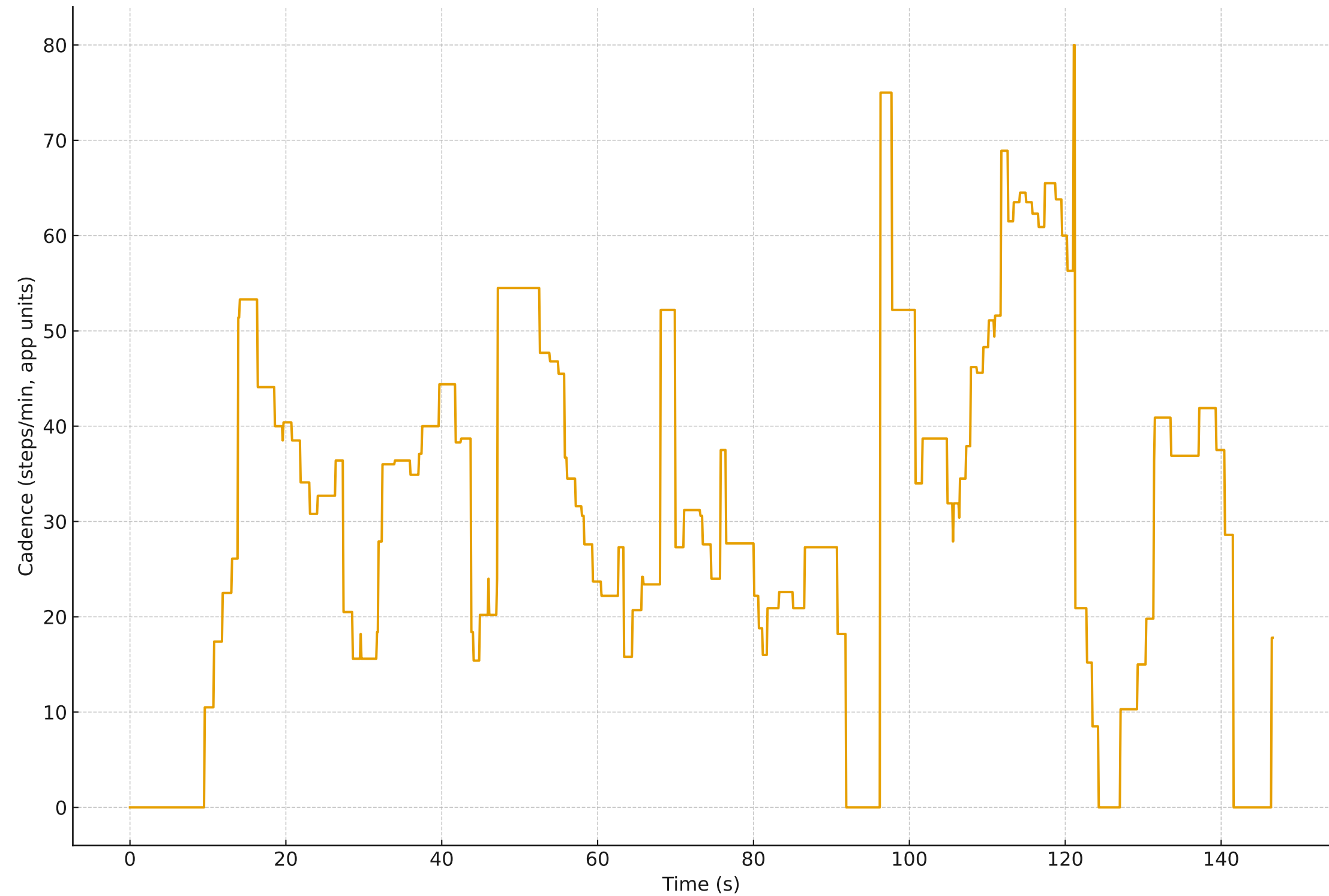
3. Step Pattern - Cumulative steps climb cleanly from 0 to 59 with no downward jumps, showing that step detection was stable (no double count or reset). - Steeper sections of the steps curve line up with the high-cadence burst near 110–120 s. - Flatter sections appear around the slow/micro-pause zone ($\sim$ 90 s) and again at the end, matching the cadence graph.

4. Foot Motion (Pitch) - The pitch signal shows regular oscillations between roughly 15° and 60° during steady walking, representing a normal heel-to-toe step cycle. - In the slower middle phase the oscillations are smaller and less frequent, consistent with shorter, slower steps. - During the fast burst the oscillations become more frequent and sometimes larger in amplitude, indicating a more dynamic, brisk gait. - Near the end of the session the pattern becomes irregular and then settles as you slow down and stop.

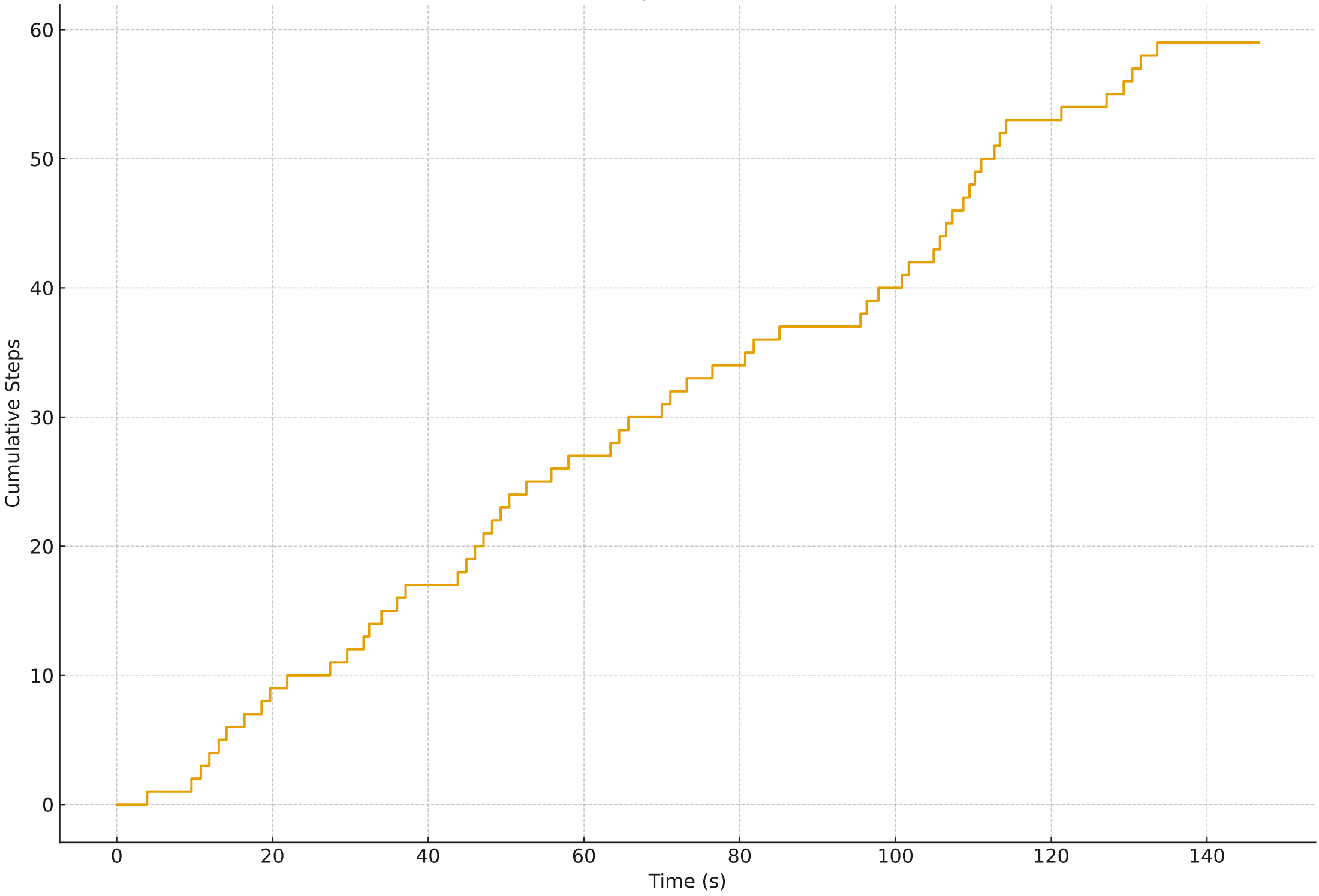
5. Temperature / Sensor Stability - Temperature drifts smoothly from about $28.0\text{--}28.1^\circ\text{C}$ down to about $25.4\text{--}25.5^\circ\text{C}$ over the session. - There are no sudden jumps, suggesting the temperature sensor is stable and the change is due to normal environmental effects or airflow while walking.

6. Overall Interpretation - You begin from rest, start walking and quickly reach a comfortable, moderate pace. - After roughly a minute you ease off a bit and there is a short micro-pause or turning phase, reflected in both cadence and step timing. - You then perform one clearly faster segment near 112–121 s, with high cadence and tightly spaced steps – this is the deliberate “walk faster” block. - Finally you reduce speed and come to a stop, giving the classic pattern: casual $\rightarrow$ slightly slower with a brief pause $\rightarrow$ brisk burst $\rightarrow$ cool-down and stop. - Step detection is clean and the strideM field in this CSV represents a single session-average stride length rather than a per-step or instantaneous value.

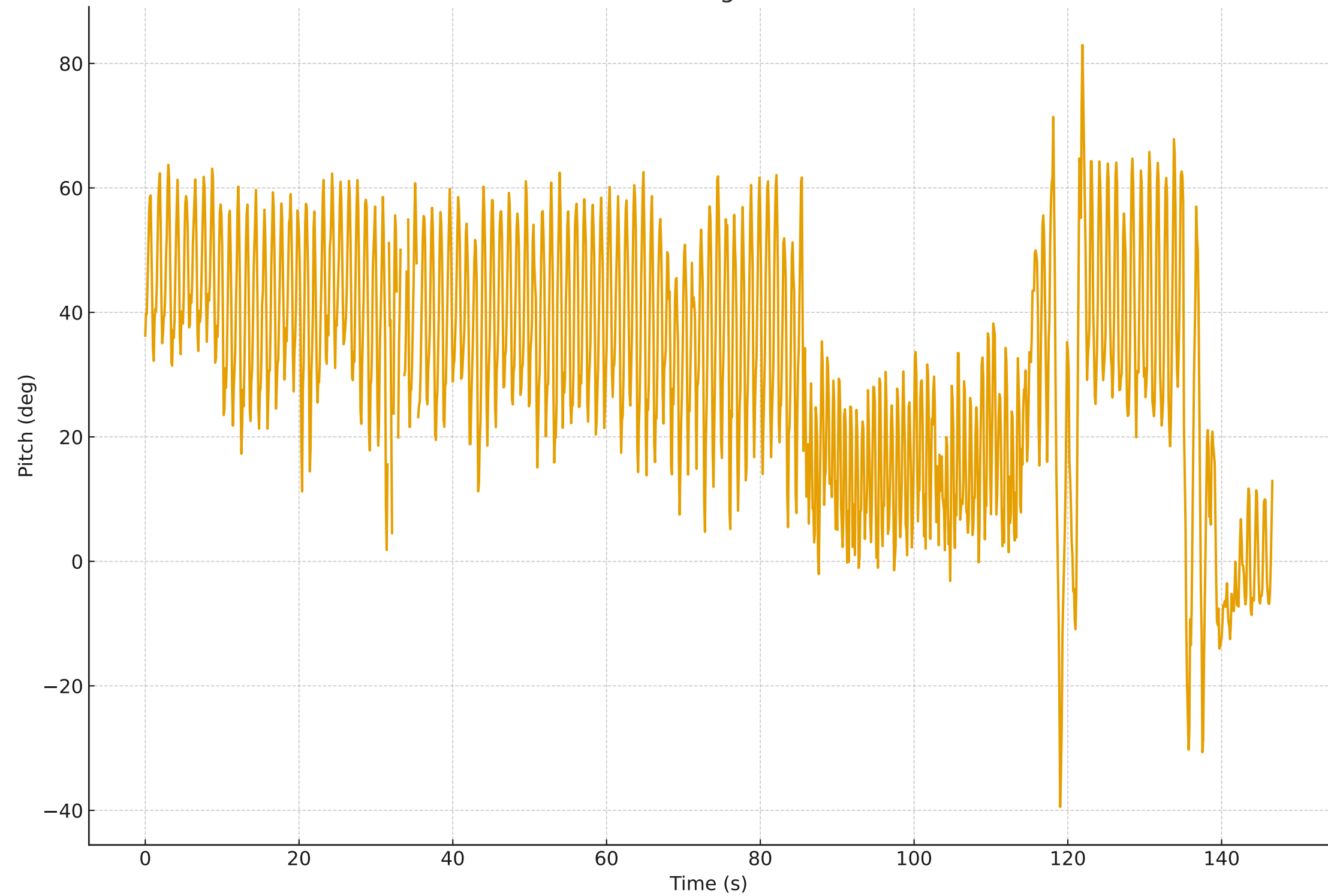
Cadence vs Time



Steps vs Time



Foot Pitch Angle vs Time



Temperature vs Time

